Next-generation Bayesian local robustness

Ryan Giordano (rgiordano@berkeley.edu, UC Berkeley)

Berkeley Neyman Seminar
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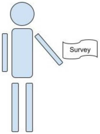
The basic problem

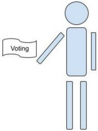
We have a survey population, for whom we observe:

- Covariates x (e.g. race, gender, zip code, age, education level)
- Responses y (e.g. A binary response to “do you support candidate Z”)

We want the average response in a target population, in which we observe only covariates.



Observe (x_i, y_i) for $i = 1, \ldots, N_S$
(This is the “Data”, $\mathcal{D}$)



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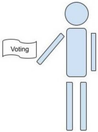

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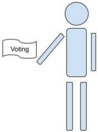

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This talk will study "Multilevel regression and poststratification" estimators:

- Set $\mathbb{E}\left[y|x, \theta\right] = m(\theta^\intercal x)$ and prior $\mathcal{P}(\theta)$ (e.g. Hierarchical logistic regression)
- Estimate the posterior $\mathcal{P}(\theta|\mathcal{D})$ with Markov Chain Monte Carlo (MCMC)
- Take $\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j$ where $\hat{y}_j = \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[y|x_j\right].$

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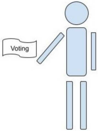

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Problem: MCMC is time consuming and the regression model is surely misspecified.
Goal: Meaningful regression specification checks with frequentist uncertainty.

Result preview

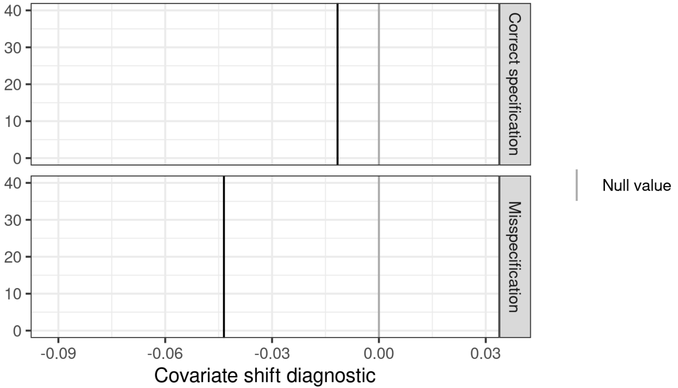

Figure 1: A covariate shift diagnostic

Here is our check for two models on the same simulated data: one correctly specified and one misspecified. The misspecied model is more extreme.

Problem: What is "large enough" to be real evidence of misspecification?
Possible answer: Frequentist confidence intervals via the parametric bootstrap.

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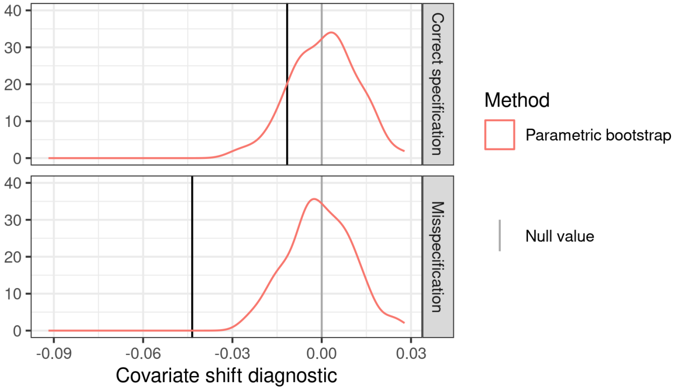

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The frequentist interval based on the parametric bootstrap rejects the misspecified model but not the correctly specified model.

Problem: Expensive: each bootstrap sample required a new MCMC run.
Possible answer: Infintesimal jackknife confidence intervals? [Giordano and Broderick, 2026]

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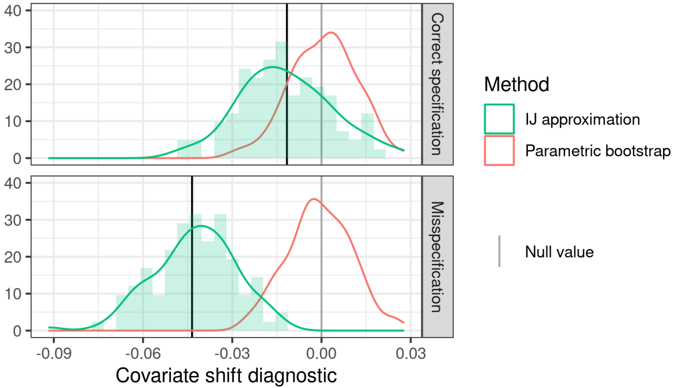

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IJ confidence intervals are fast and often work well for posterior means.
But they are not accurate for this more complex diagnostic (high Monte Carlo error).

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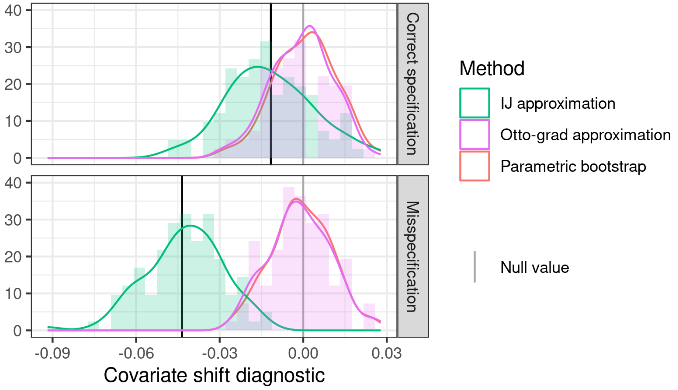

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Answer: Our flow-based local sensitivity measures ("Otto grad") produce good approximations to the bootstrap at negligible computational expense.

Outline

- Bayesian local regression specification checks
    - (With Alice Cima, Erin Hartman, Jared Miller, Avi Feller)
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
    - (With Tamara Broderick)
- Flow-based Bayesian local robustness: “Otto grad”
    - (With Lucas Schwengber)

TODO: put in pictures

Bayesian local regression specification checks

Local specification checks

$$\hat{\mu} = \frac{1}{N_T} \sum_{j=1}^{N_T} \hat{y}_j = \underbrace{\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})} \left[\frac{1}{N_T} \sum_{j=1}^{N_T} m(\theta^\intercal x_j) \right]}_{\substack{\text{Want to check specification} \\ \text{without re–running MCMC}}}.$$

Some existing local approaches:

- Specify an alternative model, maybe test with Bayes factor [Di Noia et al., Cabral et al.]
  - Bayes factors can depend strongly on priors (Lindley's paradox)
  - Articulating a family of models is inconvenient
  - Doesn't take the target into account
- Use out-of-sample checking [Vehtari et al., Kuh et al.]
  - The task is not to predict on a new IID survey datapoint

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The model is definitely misspecified, we want to check whether it's *importantly* misspecified.

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Let $\mathcal{P}(x, y)$ denote a generic joint density of x, y.
Let $\hat{\mathcal{P}}_S$ the empirical distribution on the survey data.

1. Write our estimand as a functional $\mu(\mathcal{P})$ that takes in a density and returns a MrP estimate
2. Define a covariate density perturbation $w(x)$
3. Define $\mathcal{P}_\epsilon(x, y) = \mathcal{P}(x, y) + \epsilon w(x)\mathcal{P}(y|x) = (\mathcal{P}(x) + \epsilon w(x))\,\mathcal{P}(y|x)$
4. Compute the derivative $\Delta(w, \mathcal{P}) = \partial\mu(\mathcal{P}_\epsilon)/\partial\epsilon$
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Lots of similar ideas in causal inference, kernel learning, transfer learning [Bühlmann, Rojas-Carulla et al., Muandet et al.]. This is just a local Bayesian diagnostics version.

Simple visual example

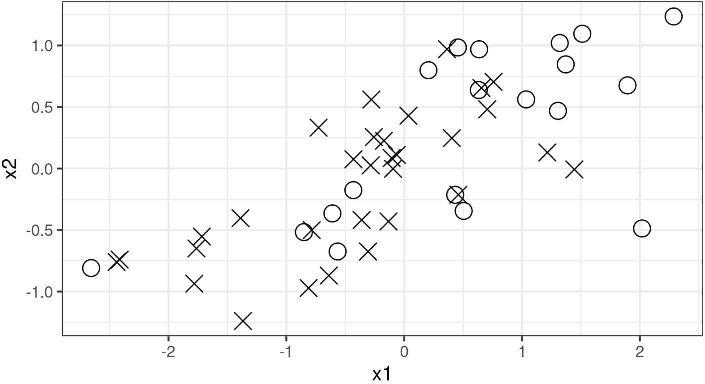

Figure 2: A covariate shift diagnostic

Consider logistic regression with two regressors, x_1 and x_2.

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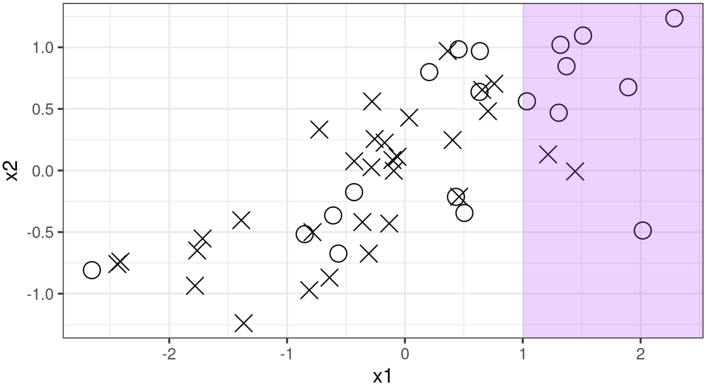

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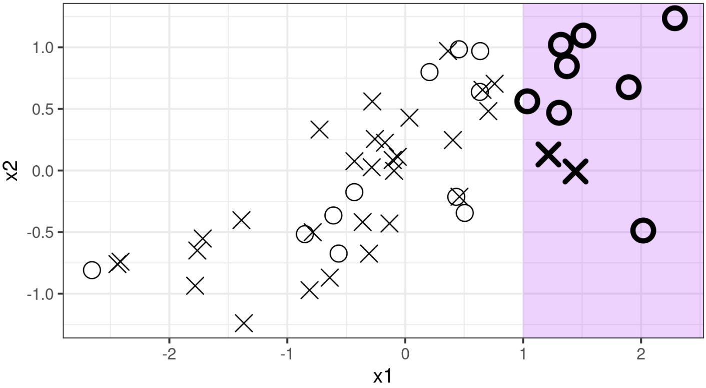

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With enough data, a regression that omits x_1 will change its coefficient of x_2 under re-weighting, whereas a correctly specified regression will not need to.

Bayesian sensitivity analysis

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right)\pi(\theta) \text{ and } \mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

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Theorem (Diaconis and Freedman, Gustafson [1996], Giordano et al. [2018]):
Consider a family of distributions $\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$ and a function $\phi(\theta)$. By interchange of integration and differentiation,

$$\left.\frac{\partial \mathbb{E}_{\mathcal{P}^{\epsilon}(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta)}\left(\phi(\theta), h(\theta)\right) .$$

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Our diagnostic is then given by

$$\Delta(\mathcal{D}) := \left.\frac{\partial \hat{\mu}}{\partial \epsilon}\right|_{\epsilon=0} = \mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\underbrace{\frac{1}{N_T}\sum_{j=1}^{N_T} m(\theta^\intercal x_j)}_{\text{Estimand}}, \underbrace{\sum_{i=1}^{N_S} w(x_i)\ell(y_i|x_i,\theta)}_{\text{Log model perturbation}}\right)$$

We expect $\Delta(w,\mathcal{D})$ to be *close* to zero when the model is correctly specified.

Bootstrapping Bayesian sensitivity analysis

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But on any actual dataset, $\Delta(w, \mathcal{D})$ will typically not be precisely zero, even under correct specification.

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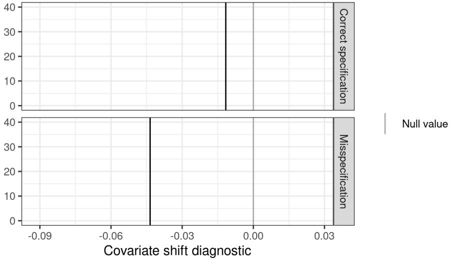

Figure 3: A covariate shift diagnostic

Simulation data: Logistic regression with five active regressors, each with covariate shift.

$$\text{Model 0 (correct)} : \texttt{y} \sim \text{Logistic}(\texttt{1 + X1 + X2 + X3 + X4 + X5})$$

$$\text{Model 1 (misspecified)} : \texttt{y} \sim \text{Logistic}(\texttt{1 + X3 + X4 + X5})$$

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To assess "closeness to zero," compare with $\Delta(\mathcal{D}^*)$, where $\mathcal{D}^*$ is a bootstrap sample.

- Compute $\Delta(\mathcal{D})$ on the original dataset
- For $b = 1, \ldots, 100$:
  1. Sample $\mathcal{D}_b^* := (x_1^*, y_1^*), \ldots (x_{N_S}^*, y_{N_S}^*)$ (parametric or nonparametric bootstrap)
  2. Run MCMC to estimate $\mathcal{P}(\theta|\mathcal{D}_b^*)$ **(time consuming!)**
  3. Use the draws to estimate $\Delta(\mathcal{D}_b^*)$
- Compare Δ with distribution of $\Delta(\mathcal{D}_b^*)$.

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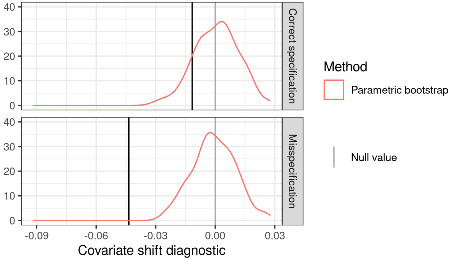

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Frequentist variance with the infinitesimal jackknife

Why frequentist variance?

For the remainder of the talk, focus on *fast approximations* to the *frequentist* sampling distribution under $x_i, y_i \overset{iid}{\sim} \mathcal{P}_S(x, y)$ of Bayesian posterior quantities like our diagnostic.

We will form *local approximations* computable only from *the original posterior*.

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We will form *local approximations* computable only from *the original posterior*.

No need to commit to our covariate shift diagnostic to care!

- Frequentist variablity of *any* model checking diagnostic is interesting
- Frequentist variablity remains meaningful under model misspecification [Huggins and Miller, 2023, Wang et al., 2026, Luo and Ji, 2026, Ji et al., 2025]

We will compare with computationally expensive full bootstrap.

Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

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We can form a series approximation:

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$$=$$

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Data re-weighting (again)

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right)\pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

If we let $w(x_i)$ denote a *bootstrap weight*, then $\mathcal{P}^\epsilon$ at $\epsilon = 1$ is the bootstrapped posterior!

We can form a series approximation:

$$\begin{aligned}
\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D}^*)}\left[\phi(\theta)\right] &\approx \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0}(1-0) \\
&= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \sum_{i=1}^{N_S} \underbrace{\left.\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right]}{\partial w_i}\right|_{w_i=0}}_{\text{"Empirical influence function" } \frac{\psi_i}{N_S}} w_i \\
&= \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] + \frac{1}{N_S}\sum_{i=1}^{N_S} \psi_i w_i
\end{aligned}$$

If this series is accurate, then we can compute ψ_i using the original posterior and quickly approximate bootstrap draws.

Bayesian von–Mises Expansion

Define the "generalized posterior" functional [Giordano and Broderick, 2026]:

$$T(\mathcal{P}, N) := \frac{\int \phi(\theta) \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right) \pi(\theta)d\theta}{\int \exp\left(N \int \ell(y|x,\theta)\mathcal{P}(x,y)dx\,dy\right) \pi(\theta)d\theta}.$$

Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

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Note that $\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] = T(\hat{\mathcal{P}}_S, N_S)$

To derive frequentist variance estimates, we study the *von Mises expansion* [von Mises, 1947]. Let $\tilde{\mathcal{P}}^t = t\hat{\mathcal{P}}_S + (1-t)\mathcal{P}_S$. Then form the series expansion in t:

$$\begin{aligned}
\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) &= \sqrt{N_S}\left(T(\hat{\mathcal{P}}_S, N_S) - T(\mathcal{P}_S, N_S)\right) \\
&= \sqrt{N_S}\left.\frac{\partial T(\tilde{\mathcal{P}}^t, N_S)}{\partial t}\right|_{t=0} (1-0) + \mathcal{E}(\tilde{t}) \quad (\tilde{t} \in [0,1]) \\
&= \underbrace{\sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi})}_{\text{Infinitesimal jackknife estimator}} + o_p(1) + \mathcal{E}(\tilde{t}).
\end{aligned}$$

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

Theorem 3 [Giordano and Broderick, 2026] (sketch):
Under Bernstein-von Mises-like conditions, $\sup_{\tilde{t}\in[0,1]} |\mathcal{E}(\tilde{t})| \xrightarrow{p} 0$.

Corollary [von Mises, 1947]: Let $\frac{1}{N_S}\sum_{i=1}^{N_S}(\psi_i - \overline{\psi})^2 \xrightarrow{p} V$. By the CLT applied to ψ_i,

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) \xrightarrow{d} \mathcal{N}\left(0, V\right).$$

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D})\exp\left(\sum_{i=1}^{N_S}\epsilon\left(\ell(y_i^*|x_i,\theta) - \ell(y_i|x_i,\theta)\right)\right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

Bayesian von–Mises Expansion Results

$$\sqrt{N_S}\left(\mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}\left[\phi(\theta)\right] - T(\mathcal{P}_S, N_S)\right) = \sqrt{N_S}\sum_{n=1}^{N}(\psi_i - \overline{\psi}) + o_p(1) + \mathcal{E}(\tilde{t}).$$

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To compute and study the parametric bootstrap instead is a simple change:

$$\mathcal{P}^{\epsilon}(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon\left(\ell(y_i^*|x_i,\theta) - \ell(y_i|x_i,\theta)\right)\right).$$

(See, e.g., Giordano et al. [2026] Theorem 4.1).

Note: The von-Mises expansion is stronger than necessary for consistency [Giordano and Broderick, 2026]. But it can be used to prove *lower bounds* on $\mathcal{E}(\tilde{t})$ and so *inconsistency* when no BVM theorem holds.

Data re-weighting.

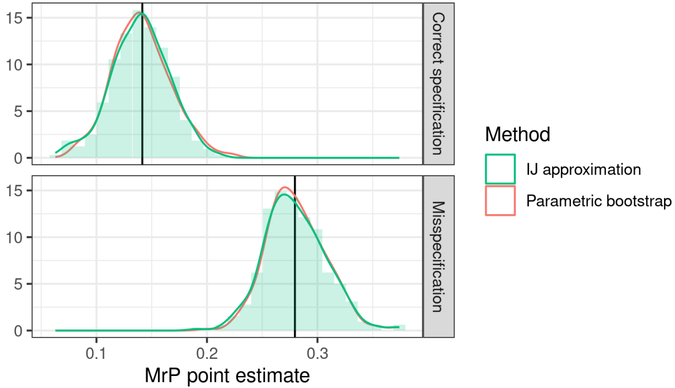

Figure 4: MrP Posterior Mean

The IJ does great for the MrP point estimate. See also Luo and Ji [2026], Ji et al. [2025].

Data re-weighting.

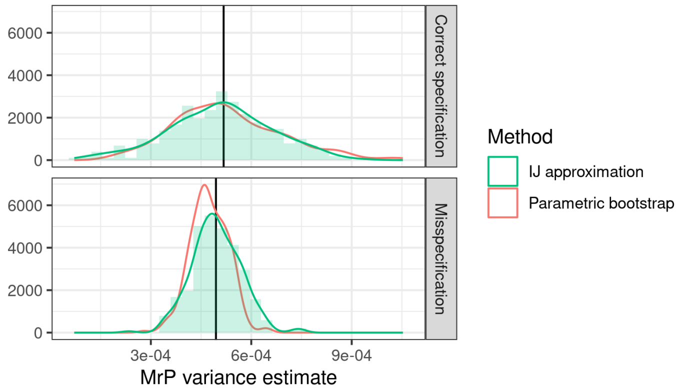

Figure 4: MrP Posterior Variance

The IJ does pretty well for the MrP posterior variance.

Data re-weighting.

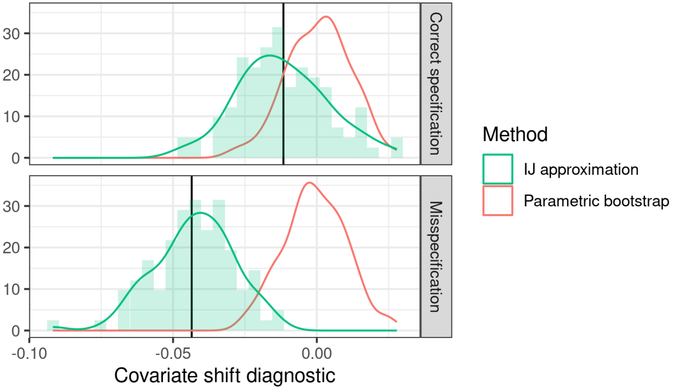

Figure 4: Covariate Shift Diagnostic

Need something better than the IJ for noisier / more complex posterior quantities!

Local robustness with flows

Why flows for sensitivity analysis?

Current Bayesian local robustness is built around linearizing posterior expectations:

$$\mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right] \approx \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta)\right] + \left.\frac{\partial \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right]}{\partial \epsilon}\right|_{\epsilon=0} (\epsilon - 0),$$

where the derivative is estimated — with noise — from MCMC draws [Gustafson, 1996, Giordano et al., 2018, Cabral et al., Di Noia et al., Giordano et al., 2026].

This has some shortcomings:

- Potential MCMC variance, not obvious how to smooth
- Linear extrapolation can lead to impossible values (e.g. negative variances)
- Complex quantities like posterior quantiles are hard to linearize
- Bayesians like to work with draws, not analytic expectations (CITE)

Idea: Instead of linearizing expectations, learn a flow from $\mathcal{P}$ to $\mathcal{P}^\epsilon$.

There's a large literature on learning flows between distant distributions. (CITE)
Our problem is easier:

- Only need "local" flows
- Our perturbations have a lot of structure (e.g. bootstraps)

Flow

Consider $\mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta)\exp(\epsilon h(\theta))$.

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ, $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

Suppose there is a scalar-valued potential function $v(\theta)$ such that

$$f^\epsilon(\theta) = \theta + \epsilon \nabla v(\theta) + O(\epsilon^2) \quad \text{as } \epsilon \to 0. \text{ Then:}$$

$$\begin{aligned}
\mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right] &= \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(f^\epsilon(\theta))\right] \Rightarrow \\
\left.\frac{\partial}{\partial \epsilon}\right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}^\epsilon(\theta)}\left[\phi(\theta)\right] &= \left.\frac{\partial}{\partial \epsilon}\right|_{\epsilon=0} \mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta + \epsilon \nabla v(\theta))\right] \\
\| \qquad\qquad & \qquad\qquad \| \\
\mathbb{E}_{\mathcal{P}(\theta)}\left[\phi(\theta) h(\theta)\right] &= \mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla \phi(\theta)^\intercal \nabla v(\theta)\right]
\end{aligned}$$

- When does such a $v(\theta)$ exist?
- How can we find it in practice?

When does such a $v(\theta)$ exist?

Let the regression log likelihood be $\ell(y|x,\theta)$, and fix $w(x)$. Then define

$$\mathcal{P}(\theta|\mathcal{D}) \propto \exp\left(\sum_{i=1}^{N_S} \ell(y_i|x_i,\theta)\right)\pi(\theta) \text{ and } \mathcal{P}^\epsilon(\theta) \propto \mathcal{P}(\theta|\mathcal{D}) \exp\left(\sum_{i=1}^{N_S} \epsilon w(x_i)\ell(y_i|x_i,\theta)\right)$$

We want a flow $\theta \mapsto f^\epsilon(\theta)$ such that, for small ϵ, $f^\epsilon(\theta) \sim \mathcal{P}^\epsilon(\theta)$ and $f^0(\theta) = \theta$.

When does such a $v(\theta)$ exist?

Theorem (Schewengber 2026): Assume that

- $\log \mathcal{P}(\theta) \in C^2(\mathcal{R}^d)$
- $\lim_{\theta\to\infty} \left(\frac{1}{2}\|\nabla \log \mathcal{P}(\theta)\|^2 - \Delta \log \mathcal{P}(\theta)\right) = +\infty.$
- $\mathbb{E}_{\mathcal{P}(\theta)}\left[h(\theta)^2\right] < \infty$

Then there exists a solution v to the differential equation

$$-\mathcal{L}_{\mathcal{P}}(v) := \nabla v^\intercal \nabla \log \mathcal{P} + \Delta v = h$$

such that $\|\nabla v\|_{L^2(\mathcal{P})} < \infty$, and the flow $\theta_\epsilon = \theta + \epsilon \nabla v(\theta)$ satsifes $\theta_\epsilon \sim \mathcal{P}^\epsilon(\theta)$ for small ϵ.

Note that $\mathcal{L}_{\mathcal{P}}$ is the drift operator, generator of Langevin diffusion, trace of the Stein operator.

How can we find $v(\theta)$ in practice?

Fix $h(\theta)$. Specify

- A family of real-valued test functions $\phi_n(\theta)$ for $n \in [N]$
- A family of real-valued basis functions $b_m(\theta)$ for $m \in [M]$

$$\text{Let } v(\theta; \beta) := \sum_{m=1}^{M} \beta_m b_m(\theta). \text{ Then}$$

$$\underbrace{\mathbb{E}_{\mathcal{P}(\theta)}\left[\phi_n(\theta) h(\theta)\right]}_{\text{Estimate with MCMC!}} = \sum_{m=1}^{M} \beta_m \underbrace{\mathbb{E}_{\mathcal{P}(\theta)}\left[\nabla \phi_n(\theta)^\intercal \nabla b_m(\theta)\right]}_{\text{Estimate with MCMC!}}$$

This gives N linear equations in M unknowns. Known as "Otto's calculus" in differential equations. Formally similar to numerical PDE solutions known as Galerkin methods, except we have a stochastic grid ($\theta \sim \mathcal{P}(\theta)$) rather than a deterministic one.

Unlike standard linear regression:

- We are free to choose ϕ_n and b_m
- Both F and B are measured with noise
- We have a lot of information about the covariance of F and B

Theorem (Schewengber 2026, sketch): The solution gives the best L2 approximation to $\|\nabla v(\theta)\|_{L^2(\mathcal{P})}$ in the span of $\phi_n(\theta)$. When $\mathcal{P}$ is Gaussian, the ordered Hermite polynomials give the best finite-rank approximation.

An elegant flow

Suppose we believe that $\mathcal{P}(\theta|\mathcal{D})$ is approximately Gaussian.

Take the basis functions $b_m(\theta) = \theta_m$ (1st Hermite polynomial) and test functions $\phi_n(\theta) = b_m(\theta)$. Then the optimal flow is given by

$$\theta_\epsilon = \theta + \epsilon \underbrace{\mathrm{Cov}_{\mathcal{P}(\theta|\mathcal{D})}\left(\theta, h(\theta)\right)}_{\left.\frac{\partial \mathbb{E}_{\mathcal{P}(\theta|\mathcal{D})}[\theta]}{\partial \epsilon}\right|_{\epsilon=0}}$$

This just translates the posterior in the direction of the mean changes.

Suppose we have $\theta_m \sim \mathcal{P}(\theta|\mathcal{D})$, and bootstrap weights w_i. Set

$$\theta_m^* = \theta_m + \frac{1}{N_S}\sum_{i=1}^{N_S} w_i \psi_i$$

and compute any posterior quantities from these samples.

Data re-weighting.

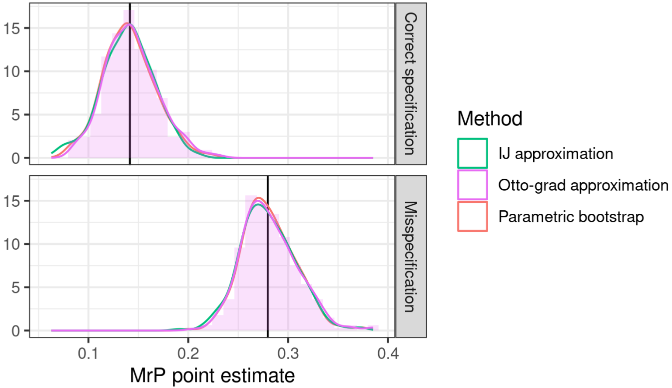

Figure 5: MrP Posterior Mean

Data re-weighting.

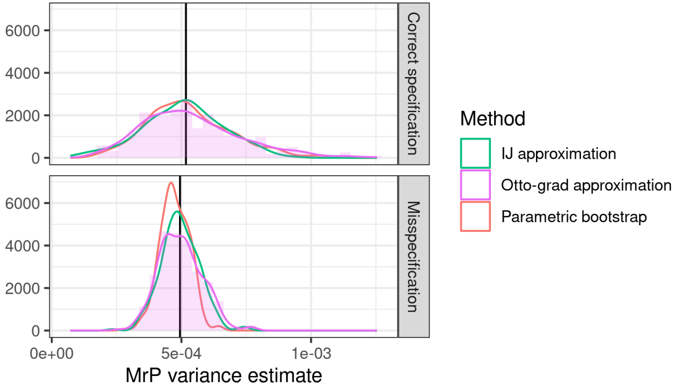

Figure 5: MrP Posterior Variance

Data re-weighting.

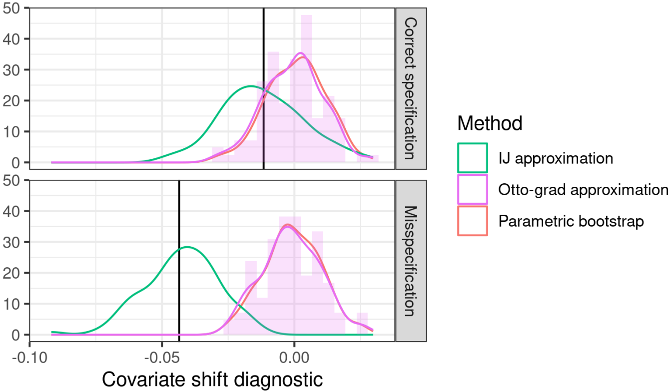

Figure 5: Covariate Shift Diagnostic

Understanding the gap between IJ and Otto is ongoing work!

Conclusion

- Bayesian local regression specification checks
  - Use invariance to covariate shift as a diagnostic for Bayesian hierarchical models
- Frequentist variability of Bayesian posteriors with the inifinitesimal jackknife
  - Use sensitivity of mean parameters to speed up bootstrapping MCMC
- Flow-based Bayesian local robustness: “Otto grad”
  - Use easy-to-compute approximate flows to speed up bootstrapping MCMC

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